



S.No: B304088

PART-I



Programe : B.Tech

Hall Ticket No: 24B11AI237

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Examination : III SEMESTER END EXAMINATIONS
REGULAR

Month-Year : Nov-25

Branch : Artificial Intelligence &
Machine Learning

Course Code : 241MA009

Course Name : Probability & Statistics

Date of Exam : 18.11.2025



Signature of the Controller of Examination

Signature of the Student with date

Signature of the Invigilator with date

Serial No.
Last Page Written

19

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
 - i. Writing their Hall Ticket No. and name in any part of the answer booklet.
 - ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
 - iii. Writing religious symbols.
 - iv. Either seeking or providing any assistance to the fellow students in the examinations.
 - v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
 - vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
 - vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**



SPACE FOR ROUGH WORK

$$f(\omega) = 2, f(s) = 3, f(g) = 3$$

$$S(\omega) = 2, S(s) = 3, S(g) = 1 \quad S(3 \text{ balls} \rightarrow g)$$

$$T(\omega) = 3, T(s) = 1, T(g) = 2 \quad P(D/s) = 3/6$$

$$f(s/6) = P(D/f)$$

$$T(4/6) = P(D/T)$$

$$P(s) = \frac{1}{6} \Rightarrow 0.8$$

$$P(f) = \frac{1}{6} \Rightarrow 0.8$$

$$P(g) = \frac{1}{6} \Rightarrow 0.8$$

P/

$$P(s|D) = \frac{P(D/s)P(s)}{P(D/f)P(f) + P(D/s)P(s) + P(D/g)P(g)}$$

$$\Rightarrow (0.5)(0.8)$$

$$\dots = (0.8)(0.8) + (0.5)(0.8) + (0.6)(0.8)$$

$$\Rightarrow 0.4$$

$$\frac{0.4}{0.64 + 0.4 + 0.48} \Rightarrow \frac{0.4}{1.52} \Rightarrow 0.26$$

$$(i) \frac{x-\mu}{\sigma} \rightarrow \text{normal } x$$

$$(ii) \frac{e^{-\lambda} \lambda^n}{n!} \rightarrow \text{poisson } x$$

$$(iii) \frac{n!}{k! (n-k)!} p^k q^{n-k} \rightarrow \text{binomial } x$$

30/

$$\frac{1}{x} = \frac{1}{10}$$

$$\int_0^{\infty} \left(\frac{1}{10}\right) e^{-\left(\frac{1}{10}\right)x} dx$$

$$\rightarrow \frac{1}{10} \left[\frac{e^{-\left(\frac{1}{10} \times 32\right)}}{e^{-\frac{1}{10}}} \right]$$

$$\Rightarrow \frac{1/12}{\frac{5}{36} + 1/12 + \frac{1}{9}}$$

$$\Rightarrow \frac{1}{12}$$

$$\frac{5}{36} + \frac{1}{12} + \frac{1}{9}$$

$$\Rightarrow \frac{0.08}{0.1 + 0.08 + 0.1}$$

$$\Rightarrow 0.208$$

$$\Rightarrow$$

$$\Rightarrow$$



UNIT-I

1b.	$x :$	0	1	2	3	4	5	6	7
	$p(x) :$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$
		$0, k$							

We know that sum of probability = 1.

$$0+k+2k+2k+3k+k^2+2k^2+7k^2+k=1.$$

$$9k+10k^2=1.$$

$$10k^2+9k-1=0.$$

$$k=0.1.$$

i), $p(x < 6)$

$$p(x=0)+p(x=1)+p(x=2)+p(x=3)+p(x=4)+p(x=5) \\ + p(x=6)$$

$$0+k+2k+2k+3k+k^2$$

$$8k+k^2$$

$$8(0.1)+(0.1)^2$$

$$=0.8+0.01$$

$$=0.81.$$

ii), $p(0 < x < 5)$

$$p(x=1)+p(x=2)+p(x=3)+p(x=4)+p(x=5)$$

$$k+2k+2k+3k+k^2$$

$$=8k$$

$$=8(0.1)$$

$$=0.8.$$

$$(iv) \text{ mean} = E(x) = 0(0) + 1(k) + 2(2k) + 3(2k) + 4(3k) + 5(k^2) + 6(2k^2) + 7(7k^2+k)$$

$$= k + 4k + 6k + 12k + 5k^2 + 12k^2 + 49k^2 + 7k$$

$$= 30k + 66k^2 \Rightarrow 30(0.1) + 66(0.1)^2$$



$$= 3 + 66(0.01)$$

$$= 0.66$$

(vi) Variance

$$= 0(0) + 1^2(K) + 2^2(K) + 3^2(2K) + 4^2(3K) + 5^2(K^2) + 6^2(K^2) +$$

$$= 0 + K + 4(2K) + 9(2K) + 16(3K) + 25(K^2) + 36(K^2) +$$

$$= 0 + K + 8K + 18K + 48K + 25K^2 + 36K^2 + 49K^2$$

$$= 124K + 112K^2$$

UNIT-II

3b). 20% of items produced from a factory are defective. find the probability in a sample of 5 chosen at random.

i) none

$$n = 20\%$$

$$\text{that means } 20/100 = 0.2$$

$$\text{So } n = 0.2 \times 100 = 20$$

p is the probability

$$p = 0.2$$

$$P(X=0) = (0.2)^0 (0.8)^5$$

$$= (1)(1) + (5)(0.2) + (10)(0.04) + (10)(0.08) + (5)(0.16) + (1)(0.32) = 1.00000$$

$$(1)(1) + (5)(0.2) + (10)(0.04) + (10)(0.08) + (5)(0.16) + (1)(0.32) = 1.00000$$

$$= 1.00000$$



ii, one is defective
 $p(x=1) =$

iii, at most 2 defective
 $p(x=0) + p(x=1) + p(x=2).$



UNIT-III

5a) various data types and the methods of data collection.

Explanation for data types:-

data types are broadly divided into two types.

1) Quantitative data.

2) Qualitative data.

Now explanation for Quantitative data.

Quantitative data:

Quantitative data is the countable Quantitative data is like average age, marks and tables and numbers. Quantitative data is countable between the ranges in discrete.

Examples are:-

marks, age, years etc.

It has two types.

1) Discrete

2) continuous.

1) ~~Discrete~~ Discrete.

Discrete is the countable ~~between the~~ ranges values.

like the values are given

it can countable.



continuous :-

continuous is the number are counted between the ranges.

Quantitative data :-

Quantitative data is the non-numerical

@ Quantitative data is the usually exact categories.

Quantitative data is the like a eyes, bag, books etc.

it has two types of orders:

1) Normal order

2) Ordinary order.

1) Normal order :-

Normal order is the it is won't have a specific order

2) Ordinary order

ordinary order have a specific order.

@ the methods of data collection :-

the methods of data collections are two methods.

they are : 1) primary data collection

2) secondary data collection.



1) primary data collection:

primary data collection is the data is collected from the source for the first time of the researchers.

Examples: Questionnaires, services, interviews and Asking question etc.

Questionnaire:

they ask question to the peoples and take the answers and after they can use.

Services:

they directly interacted with the peoples and then asking questions and take the ~~result~~ data.

Interviews:

In interviews the HR asks the question then the ~~quadrator~~ give the answer from her or him knowledge.

2) Secondary data collection:-

Secondary data is collected from the someone else by the for the ~~user~~ researcher. It is like a ~~new~~ data is collected from the newspapers and Books, and libraries.

~~news~~



(i) less than 1950 how?

$$\Rightarrow P(Z < 2) \Rightarrow P(Z < 1)$$

$$\Rightarrow \frac{\mu - x}{\sigma} \Rightarrow \frac{1950 - 2040}{60} \Rightarrow \frac{-9}{60}$$

$$\Rightarrow \frac{-9}{60} = -1.5$$

$$\Rightarrow P(Z < 0) \Rightarrow 1 - P(Z < 1)$$

$$\Rightarrow 1 - 0.332$$

$$\Rightarrow 0.5 - 0.332$$

$$\Rightarrow 0.168$$

(ii) more than 1920 and less than 2160

$$\Rightarrow (1920 < X < 2160)$$

$$\Rightarrow \frac{\mu - x}{\sigma} \Rightarrow \frac{1920 - 2040}{60}$$

$$\Rightarrow \frac{\mu - x}{\sigma} \Rightarrow \frac{2160 - 2040}{60}$$

$$\Rightarrow Z_1 = -2$$

$$\Rightarrow Z_2 = 2$$

$$\Rightarrow P(Z_1 < X < Z_2) = P(Z_1) + P(Z_2)$$

$$\Rightarrow P(-2) + P(2)$$

$$\Rightarrow 0.054 + 0.054$$

$$\Rightarrow 0.108$$



(5) (a) data types and the methods

The data types have the various types of data, that are specified in the different types of data collection.

(1) data types:-(i) Qualitative data:-

It is not numerical value. It takes the behaviours, and objects of the data. Ex: Gender, (F/M) etc.

(ii) Quantitative data:-

It takes the numerical values. It counts the whole number only.
Ex: calculation, equations.

There are two types:-

(i) discrete method:-

which is theory based method.

(ii) continuous method:-

which we can only multiple or count with whole numbers.

(2) Source of data types:-(i) primary data

In this first hand, original is find by researcher, which is original and fresh.

(i) Secondary data:-

This data is, where taken by the others which is used, before taken the secondary data.

(3) methods of data collection:-

- (i) Observation data:- where we take the value from ~~watch~~ watching like traffic light.
- (ii) Experimental data:- where the data is taken from the by practical by doing the experiments like lab.
- (iii) Schedule method:- By doing in the particular time.
- (iv) direct method:- Directly taken from the respondent.
- (v) Interview method:- Indirectly taken from the third parties like friend/witness
- (vi) indirect method:- $\uparrow \downarrow$
Written by the respondent.



(b) with replacement

The population consists of five numbers
2, 3, 6, 8, and 11

The size of sample is 2.

$$\Rightarrow N = 5$$

$$\Rightarrow n = 2$$

(i) mean of the population.

Given sample population is :-

2, 3, 6, 8, 11

$$\text{mean } (\mu) = \frac{2+3+6+8+11}{5}$$

$$\mu = \frac{30}{5}$$

$$\mu = 6$$

(ii) standard deviation of the population

$$\text{standard deviation } \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\Rightarrow \frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{5}$$

$$\Rightarrow \frac{(-2)^2 + (-3)^2 + (0)^2 + 2^2 + 5^2}{5}$$

$$\Rightarrow 16.8 \Rightarrow \sigma^2 = 16.8$$

$$\sigma = \sqrt{16.8}$$

$$\sigma = 4.1$$



mean of the sampling distribution of means,

$$2) N^n = (5)^2 = 25$$

$$2) (2,2) (2,3) (2,6) (2,8) (2,11)$$

$$= (3,2) (3,3) (3,6) (3,8) (3,11)$$

$$(6,2) (6,3) (6,6) (6,8) (6,11)$$

$$(8,2) (8,3) (8,6) (8,8) (8,11)$$

$$(11,2) (11,3) (11,6) (11,8) (11,11)$$

2) mean of sampling data

$$2) 2 \quad 4 \quad 7 \quad 9 \quad 12$$

$$4 \quad 4.5 \quad 7.5 \quad 8.2 \quad 12.5$$

$$7 \quad 7.5 \quad 9 \quad 11 \quad 14$$

$$9 \quad 8.2 \quad 11 \quad 12 \quad 15$$

$$12 \quad 12.5 \quad 14 \quad 15 \quad 16.5$$

$$2) \bar{x} = \frac{(2+4+7+9+12+14+4.5+7.5+9+11+12+12.5+14+15+16.5)}{25}$$

$$2) \frac{245.9}{25}$$

$$(9v) \quad \bar{x} = 9.8$$

$$2) \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$



$$s^2 = \frac{(2-9.8)^2 + (4-9.8)^2 + (7-9.8)^2 + \dots + (16.5-9.8)^2}{25}$$

$$\sigma^2 = \frac{\sum ni - \bar{x}}{N}$$

$$\Rightarrow \sigma^2 = \frac{60.84 + 33.64 + \dots + 27.04 + 484}{25}$$

$$\sigma^2 = 148.78$$

$$\sigma = \sqrt{148.78}$$

$$\sigma = 12.2$$

(7) (a) (i) the random sample of 49 toys
mean $\bar{x} = 49$

(ii) the mean life $\bar{x} = 15200$

(iii) the standard deviation $\sigma = 120$ km

(iv) population of mean is 15150

the to test the significance of at 0.05 level

$$\Rightarrow \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{15200 - 15150}{\frac{120}{\sqrt{49}}}$$

$$\Rightarrow \frac{50}{\frac{120}{7}} = \frac{50 \times 7}{120}$$

$$\Rightarrow \frac{350}{120} = 2.9$$

The calculated value is 2.9.



test the significant at 0.05 level is

$$n_{obs} \Rightarrow 7.815 \cdot 4.984$$

the significant at 0.05 is $\Rightarrow 7.815$
4.984

Null hypothesis is $\mu = \mu_0$

Alt hypothesis is $\mu \neq \mu_0$

the significant value at 0.05%

Null $\rightarrow$ Not null

So, Null is ^{not} hypotheses accepted

(b) The sample of the height of 6400

the englishmen has a mean 67.85

standard deviation 2.56

sample height = 1600

mean = 68.55

standard deviation = 2.52

(use α as 0.01)

$$\Rightarrow \frac{x_1 - x_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\Rightarrow n = 6400, x_1 = 67.85$$

$$n_2 = 1600, x_2 = 68.55$$

$$\sigma_1 = 2.56$$

$$\sigma_2 = 2.52$$

substituting the value of

$$\Rightarrow \frac{x_1 - x_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\Rightarrow \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} + \frac{(2.52)^2}{1600}}}$$



$$\Rightarrow \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} + \frac{(2.52)^2}{1600}}}$$

$$\Rightarrow \frac{-0.7}{\sqrt{\frac{6.55}{6400} + \frac{6.35}{1600}}}$$

$$\Rightarrow \frac{-0.7}{0.0314} \Rightarrow \frac{-0.7}{0.07} \Rightarrow -10$$

(use of α as 0.01)

$$\Rightarrow |t_{\text{calc}}| = 1 + 10 = 10$$

$$\Rightarrow (\text{use of } \alpha = 0.01) \text{ in } 10$$

$\Rightarrow$ The significant of 0.01 in 10 is 23.21 //

(10) (a) Find Karl Pearson's coefficient

(Height of father) (Height of son) Rank J1 Rank J2
(X) (Y)

65	67	8	7
66	68	7	8.5
67	64	8.5	8
67	68	8.5	8.5
68	72	4	1
69	70	3	3.5
71	69	1	4
73	70	2	3.5

Rank of X value no. of 7

8.5

m_1

Rank of Y value no. of 4

3.5

m_2

8.5

m_3



$$d = x - y \quad d^2$$

x	y
-1.5	2.25
0.5	0.25
0	0
3	9
-0.5	0.25
-3	9
-1.5	2.25

hard's pearson's coefficient of correlation

$$1 - 6 \frac{\sum d^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) + \frac{1}{12} (m_3^3 - m_3)}{n(n^2 - 1)}$$

$$\Rightarrow 1 - 6 \frac{24 + \frac{1}{12} (614 - 8.5) + \frac{1}{12} (42.8 - 3.5) + \frac{1}{12} (614 - 8.5)}{8(8^2 - 1)}$$

$$\Rightarrow 1 - 6 \cdot \left(24 + \frac{605}{12} + \frac{39.3}{12} + \frac{605}{12} \right) \div 8(63)$$

$$\Rightarrow \frac{1 - 6(24 + 50.4 + 3.27 + 50.4)}{512} \Rightarrow \frac{1 - 6(128.07)}{512}$$

$$\Rightarrow \frac{1 - 768.42}{512} \Rightarrow \frac{-767.42}{512} \Rightarrow -1.498 //$$



$$(b) \quad 3x + 12y = 19 \quad \text{--- (1)}$$

$$3y + 9x = 46 \quad \text{--- (2)}$$

One equation in the question is

$$3x + 12y = 19 \quad \text{--- (1)}$$

$$3y + 9x = 46 \quad \text{--- (2)}$$

To find the 'y' value:-

$$y = \frac{\cancel{9x} + 46}{46}$$

The value 'y' is substituted in eq (1)

$$3x + 12y = 19$$

$$3x + 12y - 19 = 0$$

$$\Rightarrow 3x + 12\left(\frac{9x + 46}{46}\right) - 19 = 0$$

$$\Rightarrow \cancel{138x} + 36 + 3x + 12\left(\frac{9x + 46}{46}\right) - 19 = 0$$

$$\Rightarrow 138x + 108x - 36 - 874 = 0$$

$$\Rightarrow \cancel{18} \cdot 246x - 910 = 0$$

$$246x = 910$$

$$x = \frac{910}{246} \approx 3.6$$

Substituting the value of 'x' in eq (2)

$$= 3y + 9x - 46 = 0$$

$$\Rightarrow 3y + 9(3.6) - 46 = 0$$

$$\Rightarrow 3y + 32 - 46 = 0$$

$$\Rightarrow 3y + 13 = 0$$

$$3y = 13 \Rightarrow y = 13/3$$



$$(x = 3.6 ; y = 13/3)$$

2) from eq (1)

$$3x + 12y - 19 = 0$$

$$\Rightarrow b_{xy} = 3/12$$

$$\Rightarrow 3y + 9x - 46 = 0$$

$$\Rightarrow b_{yx} = 3/9$$

(i) The value of correlation coefficient

$$\Rightarrow \sqrt{b_{yx} \times b_{xy}}$$

$$\Rightarrow \sqrt{(3/12) \times (3/9)}$$

$$\Rightarrow \sqrt{(1/4) \times (1/3)}$$

$$\Rightarrow \sqrt{\frac{4 \times 3}{12}} \Rightarrow \frac{\sqrt{3}}{6}$$

$$\sqrt{1/12}$$

$$\sqrt{1/12}$$

(ii) The value of mean of x and y

$$\Rightarrow \frac{3.6 + 13/3}{2}$$

$$\Rightarrow \frac{7.9}{2} \Rightarrow 3.9$$

The mean of x and y is 3.9,,



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